INCH-POUND

MIL-PRF-44469A
26 July 1996
SUPERSEDING
MIL-C-44469
28 May 1993

PERFORMANCE SPECIFICATION

CHILI AND MACARONI, PACKAGED IN A FLEXIBLE POUCH, SHELF STABLE

This specification is approved for usc by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance criteria for production of a finished product consisting of shelf stable chili and macaroni, packaged in a flexible pouch, suitable for use by the Department of Defense as a component of operational rations. This component is consumed by combat personnel under worldwide environmental extremes as part of an operational ration, and is a source of nutritional intake. It is essential that this item be produced in accordance with good commercial practice to attain high standards of appearance, odor, flavor, and texture so that high levels of troop acceptance are achieved.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 4 and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 4 and 5 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Soldier Systems Command, Natick Research, Development, and Engineering Center, ATTN: SSCNC-WRE, Natick, MA 01760-5018 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form apart of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-44073 - Packaging of Food in Flexible Pouches

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.1).

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC International

(Application for copies should be addressed to the AOAC International, 481 North Frederick Avenue, Suite 500, Gaithersburg, MD 20877.)

2.4 Order of precedencc. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.1), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Performance characteristics of the packaged food.

3.2.1 Commercial sterility. The packaged food shall be processed until commercially sterile.

3.2.2 Shelf life. The packaged food shall meet the minimum shelf life requirement of 36 months at 80°F.

3.2.3 Appearance.

3.2.3.1 Ground beef. The beef shall be a grind size typically produced by a 3/8 inch grinder plate, and shall be free of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material. The cooked ground beef shall have a characteristic cooked color.

3.2.3.2 Enriched elbow macaroni. The macaroni shall be curved strands not less than 3/4 inches or greater than 1.5 inches in length. The macaroni shall be an off white to light tan color.

3.2.3.3 Sauce. The sauce shall be a dark red to reddish brown color.

3.2.3.4 General. There shall be no bone or bone fragments in the packaged food.

3.2.4 Odor and flavor.

3.2.4.1 General. The packaged food shall have an odor and flavor characteristic of processed chili and macaroni in a spicy chili pepper blend tomato based sauce.

3.2.4.2 Foreign. The packaged food shall be free from foreign odors and flavors such as, but not limited to, burnt, scorched, rancid, sour, or stale.

3.2.5 Texture.

3.2.5.1 Enriched elbow macaroni. The macaroni shall be slightly soft to slightly firm. The macaroni shall not be pasty.

3.2.5.2 Sauce. The sauce shall be moderately thick.

3.2.6 Weight.

3.2.6.1 Net weight. The average net weight shall be not less than 8.0 ounces. No individual pouch shall contain less than 7.5 ounces.

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3.2.6.2 Drained weight.

3.2.6.2.1 Beef. The average drained weight of the ground beef shall be not less than 2.0 ounces. The drained weight of the ground beef in an individual pouch shall be not less than 1.8 ounces.

3.2.6.2.2 Enriched elbow macaroni. The drained weight of the macaroni in an individual pouch shall be not less than 2.6 ounces.

3.2.7 Palatability. The finished product shall be equal to or better than the approved first article, when applicable, or other approved model in palatability and overall appearance.

3.2.8 Nutrient content.

3.2.8.1 Water content. The water content shall be not greater than 76.7 percent.

3.2.8.2 Protein content. The protein content shall be not less than 10.8 percent.

3.2.8.3 Fat content. The fat content shall be not greater than 4.0 percent.

3.2.8.4 Ash content. The ash content pouch shall be not greater than 1.6 percent.

3.2.8.5 Salt content. The salt content shall be not greater than 1.0 percent.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article shall be inspected in accordance with the provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the performance characteristics or any appearance or palatability failure shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the methods of inspection of 4.5.1 through 4.5.8.

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4.4 Examination. The finished product shall be examined for compliance with the requirements specified in 3.2.3, 3.2.4, 3.2.5, 3.2.6, and 3.2.7. Defects and defect classifications are listed in table I.

TABLE 1. Product defects 1/ 2/

Category		Defect
Major	Minor	
		Appearance
101		Beef grind size not as specified
102		Color not characteristic of cooked ground beef
103		Bone or bone fragment measuring more than 0.3 inch in any dimension
	201	Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material is more than 0.35 ounces
	202	Color of macaroni not off white to light tan color
	203	Color of sauce not dark red to reddish brown
		Odor and flavor
104		Odor and flavor not characteristic of processed ground beef and macaroni in a spicy chili pepper blend tomato sauce
		Texture
	204	Macaroni not slightly soft to slightly firm
	205	Macaroni is pasty
	206	Sauce not moderately thick
		Weight
	207	Net weight of an individual pouch less than 7.5 ounces 3/

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TABLE I. Product defects 1/ 2/ cont'd

Category		Defect
Major	Minor	
	208	Drained weight of ground beef not as specified 4/
	209	Drained weight of macaroni not as specified

1/ Presence of any foreign odors and flavors such as, but not limited to burnt, scorched, rancid, sour, or stale shall be cause for rejection of the lot.

2/ Finished product not equal to or better than the approved first article, when applicable, or other approved model in palatability and overall appearance shall be cause for rejection of the lot.

3/ If the sample average net weight is less than 8.0 ounces, the lot shall be rejected.

4/ If the sample average drained weight of ground beef is less than 2.0 ounces, the lot shall be rejected.

4.5 Methods of inspection.

4.5.1 Commercial sterility. Testing for commercial sterility shall be in accordance with MIL-PRF-44073.

4.5.2 Shelf life. Compliance with 3.2.2 shall be determined by incubation for 1 month at 120°F or 6 months at 10O°F or 36 months at 80°F. Contractor shall provide a certificate of conformance.

4.5.3 Appearance. Appearance acceptability is determined by compliance with 4.4.

4.5.4 Odor and flavor. Odor and flavor acceptability is determined by compliance with 4.4.

4.5.5 Texture. Texture acceptability is determined by compliance with 4.4.

4.5.6 Weight.

4.5.6.1 Net weight. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Results shall be reported to the nearest 0.1 ounce.

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4.5.6.2 Drained weight. The contents of the pouch shall be poured into a flat-bottom container. A minimum of three times the volume of the pouch of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated so as to liquify rendered fat and remove the sauce. The contents shall then be poured into a U.S. Standard No. 7 sieve in a manner that will distribute the product over the sieve. The sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4.5.7 Palatability. Palatability acceptability is determined by compliance with 4.4.

4.5.8 Nutrient content. The sample to be analyzed shall be a composite of eight filled and sealed pouches which have been selected at random from the lot. The composited sample shall be prepared and analyzed for water content, protein content, fat content, ash content, and salt content in accordance with the following methods of the Official Methods of Analysis of AOAC International:

Test	Method Number
Moisture	950.46A
Protein	988.05
Fat	960.39
Ash	923.03
Salt	935.47

Test results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirements specified in 3.2.8 shall be cause for rejection of the lot.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in MIL-PRF-44073.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

c. When a first article is not required (see 3.1).

6.2 Ingredients and formulation. Ingredients and formulation may be as follows:

Sauce:	Ingredients	Percent by weight
	Beef, raw	53.00
	Water	27.32
	Tomatoes, crushed	7.50
	Tomato paste (26% solids)	5.50
	Chili powder	2.20
	Starch, waxy maize, modified	1.50
	Onion, dehydrated, chopped	1.00
	Salt	0.95
	Sugar, brown, light	0.75
	Lecithin	0.25
	Pepper, black, ground	0.03

Product:	Ingredients	Percent by weight
	Meat sauce	85.0
	Macaroni	15.0

6.3 Technical information. Specific technical inquiries may be addressed to the Commander, U.S. Army Soldier Systems Command, Natick Research, Development and Engineering Center, ATTN: SSCNC-WRE, Natick, MA 01760-5018.

6.4 Subject term (key word) listing.

Operational ration
Food processing
Pouched food

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 35

Preparing activity:

Army - GL

(Project 8940-0847)

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

Civil agency coordinating activity:

USDA-FV

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. **The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.**
2. **The submitter of this form must complete blocks 4, 5, 6, and 7.**
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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-44469	**2. DOCUMENT DATE (*YYMMDD*)** 1996 July 26

3. DOCUMENT TITLE CHILI AND MACARONI, PACKAGED IN A FLEXIBLE POUCH, SHELF STABLE

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME ***(Last, First, Middle Initial)***	**b. ORGANIZATION**	
c. ADDRESS ***(Include Zip Code)***	**d. TELEPHONE (*Include Area Code*)** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED** ***(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME U.S. Army Soldier Systems Command, Natick Research, Development, and Engineering Center	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** 508-233-5907 **(2) AUTOVON/DSN** 256-5907
c. ADDRESS *(Include Zip Code)* Commander, U.S. Army Soldier Systems Command, Natick Research, Development, and Engineering Center ATTN: SSCNC-WRE, Natick, MA 01760-5018	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Standardization Program Division** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3346** **Telephone (703) 681-9340 AUTOVON 761-9340**

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